SHAAN COOPER

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Education

Embry-Riddle Aeronautical University, Daytona Beach, FL *Expected Graduation May 2029*

Bachelor of Science Aerospace Engineering

Project Experience | Experimental Rocket Propulsion Lab (ERPL)

Liquid Bi-Propellant Feed System Redesign and Rebuild | *Feb 2026 – April 2026*

- Redesigned and rebuilt a damaged bi-propellant test stand, integrating bang-bang solenoid actuation to eliminate regulator droop and restore full testing capabilities.
- Engineered systemic improvements to withstand possible engine failures, preventing the mechanical failures and repairs associated with the previous iteration.
- Optimized system design to allow for operational procedures to reduce hot-fire setup time by over 30%, decreasing average preparation time from 6 hours to 4 hours.
- Enabled future system scalability and upgradeability while reducing operational safety risks during testing.

Next-Generation Feed System Design | *Jan 2026 – March 2026*

- Co-led the design of a next-generation propulsion feed system supporting engines up to ~10,000 lbf thrust, representing a ~200% increase over current system capability.
- Engineered expanded feed-system capabilities to accommodate cryogenic propellants, throttle control, increased propellant capacity, and higher mass flow rates.
- Optimized system architecture to improve mobility, enabling a more compact and transportable test configuration.

System and Propulsion Analysis - Liquid Bi-Propellant Feed System | *Aug 2025 – Present*

- Developed a Python-based analysis tool to model pressure losses and mass flow rates in a liquid bi-propellant feed system using Venturi flow principles, informing tank pressurization requirements and predicting target chamber pressure within ~5 psi prior to regulator droop.
- Evaluated post-hot-fire test data to characterize propulsion system performance, analyzing C*, specific impulse (Isp), thrust coefficient (Cf), system pressures, valve states, and propellant temperatures to assess engine and feed-system behavior.
- Performed finite element analysis in ANSYS Mechanical on a vertical engine test stand to verify structural integrity under potential detonation loads, supporting safe testing of a 2,500-lbf-class engine and protection of nearby vehicle fluid systems.

Feed System Maintenance & Installation - Liquid Bi-Propellant Feed System | *Aug 2025 – Present*

- Executed routine maintenance and integrated upgraded feed-system components to support reliable and safe operation under high-pressure conditions.
- Fabricated and modified fluid-system tubing through precision bending and flaring; installed pneumatic ball valves, check valves, relief valves, and various fittings while identifying potential failure points to improve system reliability and operational safety.
- Soldered and routed electrical wiring for pressure transducers and solenoid valves, supporting reliable instrumentation, data acquisition, and valve actuation.

Skills

Engineering Software: MATLAB, Python, Autodesk Inventor, Ansys, CATIA V5

- Utilized MATLAB and Python to process, visualize, and analyze propulsion test data, including pressure, temperature, thrust, specific impulse (Isp), and characteristic velocity (C*).
- Designed and modeled fluid-system components and complex fitting assemblies using CATIA V5 and Autodesk Inventor, as well as performed ANSYS structural analyses to evaluate the integrity of test-stand supports and load-bearing components.

Office Tools: Microsoft Excel, Word, PowerPoint, Visio, Teams

- Leveraged Microsoft Office tools to develop technical documentation, engineering diagrams, perform calculations and data analysis, and communicate results through presentations and meetings.

Leadership/Involvement

Experimental Rocket Propulsion Lab: Fluids Division Member

- Maintained and enhanced multiple propulsion fluid-system projects, including two ground feed systems and vehicle fluid systems, while developing analytical models to characterize system performance and behavior.